

DSA Practical Test

Name: Kalpit Salvi

Date: 11-6-26

Time: 1 Hour

Total Marks: 30

Section A — Dry Run (15 Marks)

Q1. Linear Search Trace (5 Marks)

Array = {8, 3, 17, 6, 42, 15, 29}

Target = 15

Fill in the table:

Step	Index	Value	Match?
1	0	8	No
2	1	3	No
3	2	17	No
4	3	6	No
5	4	42	No

6	5	15	Yes
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Q1a. Found at index = 5

Q1b. Total steps = 6

Q1c. Best Case / Average Case / Worst Case?
Explain why.

Ans:

Best Case = 8

As 8 is the first element in the array.

Average case = 6

As 6 is right in the middle of the array.

Worst Case = 29

As 29 is in the end of the array.

Q2. Binary Search Trace (5 Marks)

Array = {5, 11, 19, 27, 35, 43, 58}

Target = 43

Fill in the table:

Step	Left	Right	Mid	Value	Decision
1	0	6	3	27	Ignore Left side (left = mid + 1)
2	4	6	5	43	Found
3					

Q2a. Found at index = 5

Q2b. Total steps = 2

Q2c. How many steps would Linear Search take for the same target?

Ans : 6

Q3. Bubble Sort Trace (5 Marks)

Array = {45, 12, 33, 8, 27}

Complete Pass 1 and Pass 2.

Pass 1:

Step 1: 45 and 12 → Swap? Yes → Array: {12,45,33,8,27};

Step 2: 45 and 33 → Swap? Yes → Array: {12,33,45,8,27}

Step 3: 45 and 8 → Swap? Yes → Array: {12,33,8,45,27}

Step 4: 45 and 27 → Swap? → Yes Array: {12,33,8,27,45}

Which element settled after Pass 1? 45

Pass 2:

Step 1: 12 and 33 → Swap? No → Array: {12,33,8,27,45}

Step 2: 33 and 8 → Swap? Yes → Array: {12,8,33,27,45}

Step 3: 33 and 27 → Swap? Yes → Array: {12,8,27,33,45}

Which element settled after Pass 2? 33 and 45

Section B — Coding (15 Marks)

Q4. Linear Search Program (5 Marks)

Write a Linear Search program for:

Array = {23, 67, 4, 89, 31, 55, 12}

Target = 89

Requirements:

- Print each step
 - Print Found or Not Found
 - Print index if found
 - Print total steps taken
-

Q5. Binary Search Program (5 Marks)

Write a Binary Search program for:

Array = {4, 12, 23, 31, 55, 67, 89}

Target = 12

Requirements:

- Print Left, Right, Mid and Value at each step
 - Print Found or Not Found
 - Print total steps taken
-

Q6. Bubble Sort Program (5 Marks)

Write a Bubble Sort program for:

Array = {64, 25, 12, 22, 11}

Requirements:

- Print array before sorting
- Print array after every swap
- Print total swaps
- Print final sorted array

Instructions

1. Dry Run questions may be answered in a text file on your system.
2. Coding questions should be completed using separate program files on your system.

3. Save all files properly and submit them as instructed.
4. Time Duration: 1 Hour.